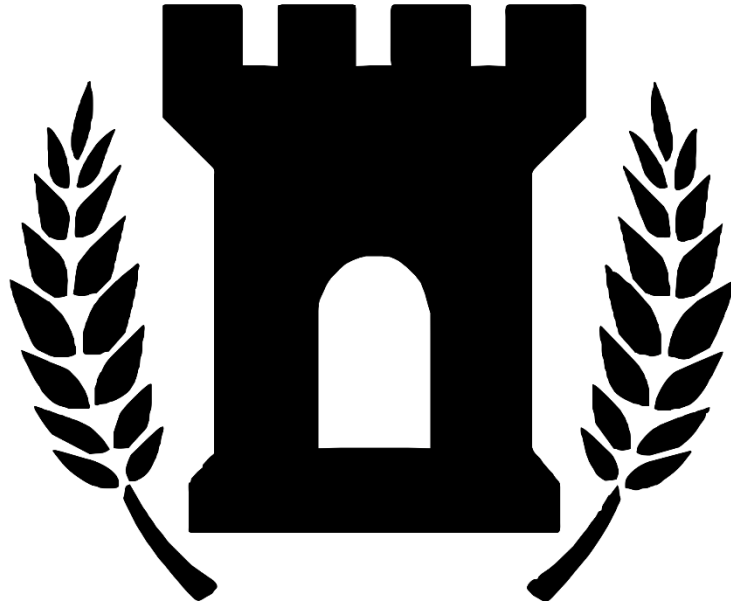




Feldfeste Operation Instruction



Feldfeste Games Studio

FF-OI-07: “Manual Level Design”

Document History

Revision	Content	Editor	Time
01	Initial Document Creation	Hannes	15.06.2026



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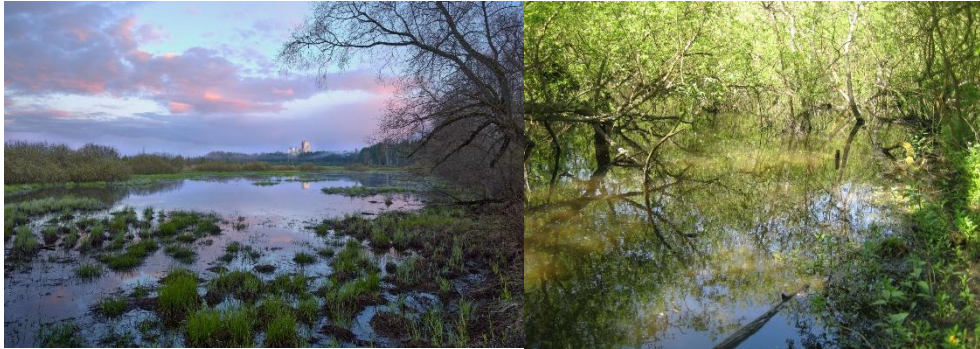
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1. Overview and Purpose

This guide is supposed to be a quick reference for any level designer in our team to build levels. Just follow the Scheme and you should be fine. As this is a work in progress, details will be updated and added. For any feedback and questions, please directly target our discord channel for **level design** questions!

The Maps we create for Castle Age share the following characteristics:

- real-world reference (use geodata as a base)
- relatively small, just a few square kilometers
- mostly around historic areas of interest like castles or battle fields
- trying to emulate nature in the environment
- trying to re-create a "naturalistic" appearance
- using one of the specific **biomes** we plan:
 - north european grassland (flat, swampy)
<https://trello.com/c/jQYtemRc/482-northern-swamp-biome-weather-picked-custom-maps-flatlands>



- west-european atlantic coast (cliffs, beach, gras)
<https://trello.com/c/um914wFR/815-western-atlantic-coast-biome-weather-custom-maps-normandy>



- Middle-European mid-mountain forests
<https://trello.com/c/YsLZ77z9/480-middle-range-mountains-blackforest-thuringian-forest-bavarian-forest-erzgebirge-etc-weather-maps>

(blackforest, france, bohemian forest, bavarian forest, thuringian forest etc)



- Alpine Biome
<https://trello.com/c/8NeOquUe/816-alpine-biome-weather-custom-maps>



- Western mediterranean: (Italy, Iberia)

<https://trello.com/c/IFZRKckJ/481-mediterranean-biome-weather-maps>



- **[DLC]** mediterranean coast (cliffs, beach, subtropical) [greek isles]

<https://trello.com/c/99ySIJC1/504-east-mediterranean-biome-greek-isles-weather-maps-animals>



- **[DLC]** AnatoliaPersia

<https://trello.com/c/IOhRsJJU/487-oriental-biome-anatolia-persia-weather-maps-animals>



- **[DLC]** Mongol Steppe
<https://trello.com/c/iJqUcM1u/493-desert-and-steppe-biome-weather-maps-animals>



- **[DLC]** North Africa



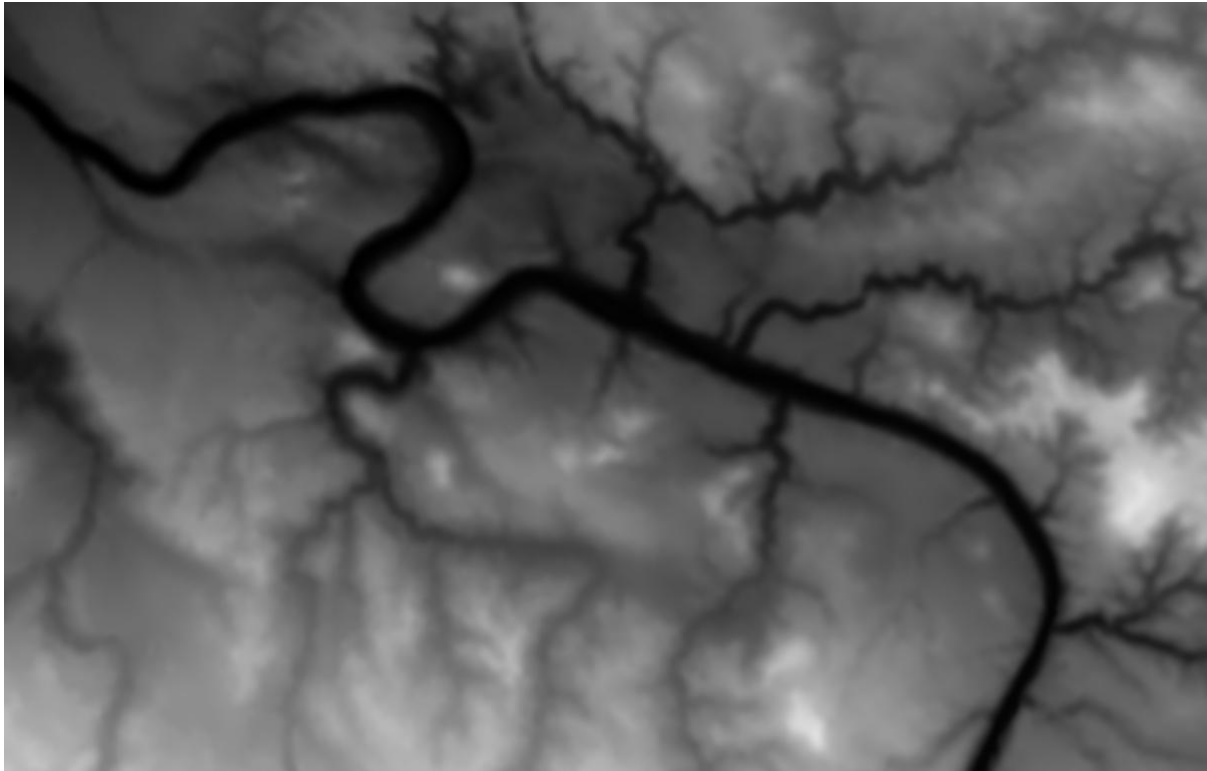
<https://trello.com/c/C7a1y4Ji/498-north-african-biome-animals-maps-weather>

- **[DLC]** Middle East /Holy Land
- Each of these have their regional:
 - Weathers
 - Animal Encounters
 - Terrain Textures
 - Foliage (Bushes, Grass)
 - Trees
 - Rocks

Overview of the Workflow

- <https://docs.unrealengine.com/4.27/en-US/BuildingWorlds/Landscape/Creation/>
- Create Landscape
- Apply Material
- Check that layer info is correct
- [Landscape Edit Layers in Unreal Engine](#)
- <https://docs.unrealengine.com/4.27/en-US/BuildingWorlds/Landscape/Editing/SculptMode/Blueprint/>
- ([Voxel Plugin](#)) can do real-time editing of landscape (not required) (Brushes for siege impacts? Destroyable terrain?)

2. Heightmap Creation



- Data Sources:
 - Extract 3d Data from Online Maps
 - Outside UE:
 - Polish and fit Map Data
 - Export in Texture Formate
 - Inside UE:
 - Import in Unreal Engine
 - Final Tweaks in Unreal Engine
-
1. get XYZ-file of the region
 2. import to CloudCompare (freeware)
 3. export mesh from cloudcompare
 4. import mesh in 3dsMax
 5. texture with gradient and render top view with 16bit
 6. use generated height map in Unreal as landscape

1.1 Extract 3d Data from Online Maps

1. **Sachsen** (in the game the Saxon eastern marches, mostly Meißen, Zeitz, Lausitz etc.)
<https://www.geodaten.sachsen.de/downloadbereich-digitale-hoehenmodelle-4851.html>
2. **Bayern** (part of Herzogtum Bayern and Nordmark)
<https://geodaten.bayern.de/opengeodata/OpenDataDetail.html?pn=dgm1>

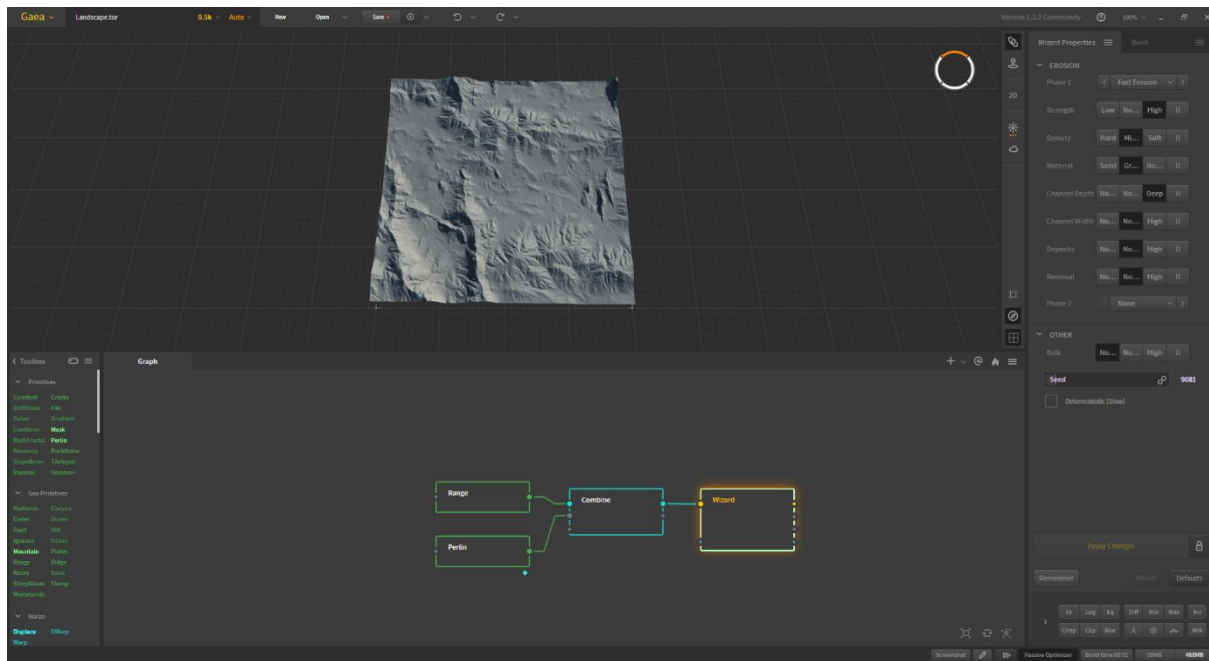
3. **Thüringen** (Thüringen) <https://www.geoportal-th.de/de-de/Downloadbereiche/Download-Offene-Geodaten-Th%C3%BCrtingen/Download-H%C3%B6hendaten>
4. **Niedersachsen** (in the game part of Herzogtum Sachsen) <https://opengeodata.lgl.niedersachsen.de/#bdlm>
5. **Tyrol** (part of Bavarian eastern marches, Austria) <https://maps.civis.bz.it/#/?context=PROV-BZ-MONUMENT-ARCHAEOLOGY&lang=de&bbox=1275156.6904,5923073.1162,1275297.7807,5923216.8152>
6. **Baden-Württemberg** (in the game part of Herzogtum Schwaben) [https://www.geoportal-bw.de/#/\(sidenav:menu\)](https://www.geoportal-bw.de/#/(sidenav:menu))
7. **Schweiz** (part of Herzogtum Schwaben in the game) <https://www.geo.admin.ch/de/geodatenmodelle>
8. **Netherlands** (not sure) https://data.overheid.nl/datasets?facet_theme%5B0%5D=http%3A//standaarden.overheid.nl/owms/terms/Natuur_en_milieu
9. **Paris** (not sure) <https://opendata.paris.fr/pages/home/>
10. **France** (need to investigate further) <https://opendata.paris.fr/pages/home/>
11. **Italy** (here Bologna example) <https://www.dati.gov.it/view-dataset?groups=ambiente&organization=comune-di-bologna>
12. **Tuscany** (Herzogtum Toskana in the game) <https://www.regione.toscana.it/-/open-geodata>
13. **Flanders/Belgium** <https://www.geo.be/home?l=de>
14. **Spain** <https://datos.gob.es/en>
15. **Normandy** (Herzogtum Normandie in the game) <https://data.maptiler.com/downloads/europe/france/basse-normandie/>
16. maybe this tool can help with many places: <https://data.maptiler.com/downloads/europe/>

1.2 Optional Polish outside the Editor

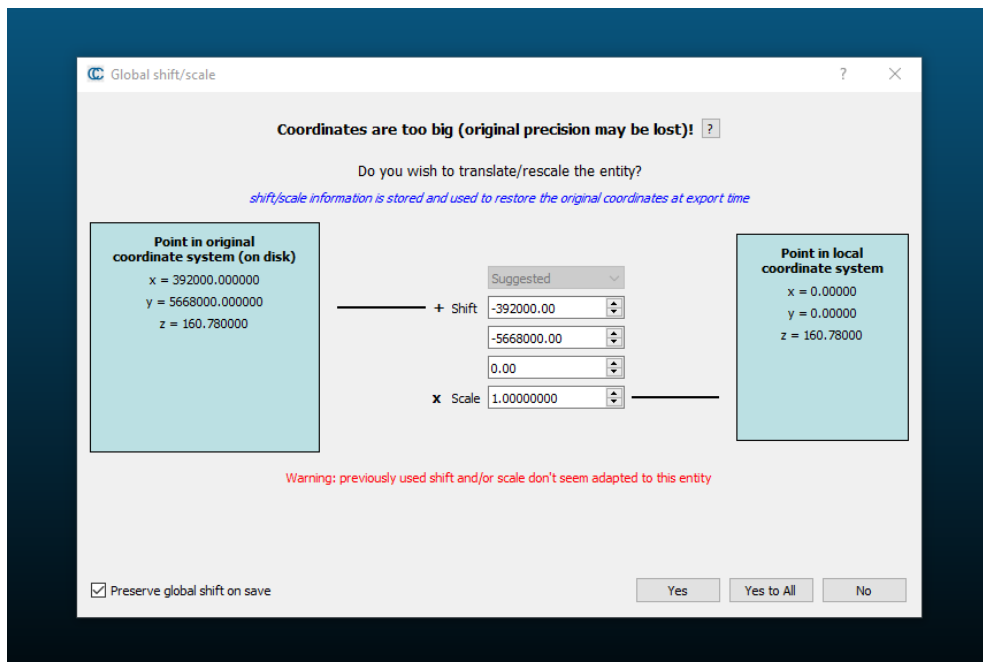
- Gaea
Problem of costs/ License
- <https://quadspinner.com/Download>

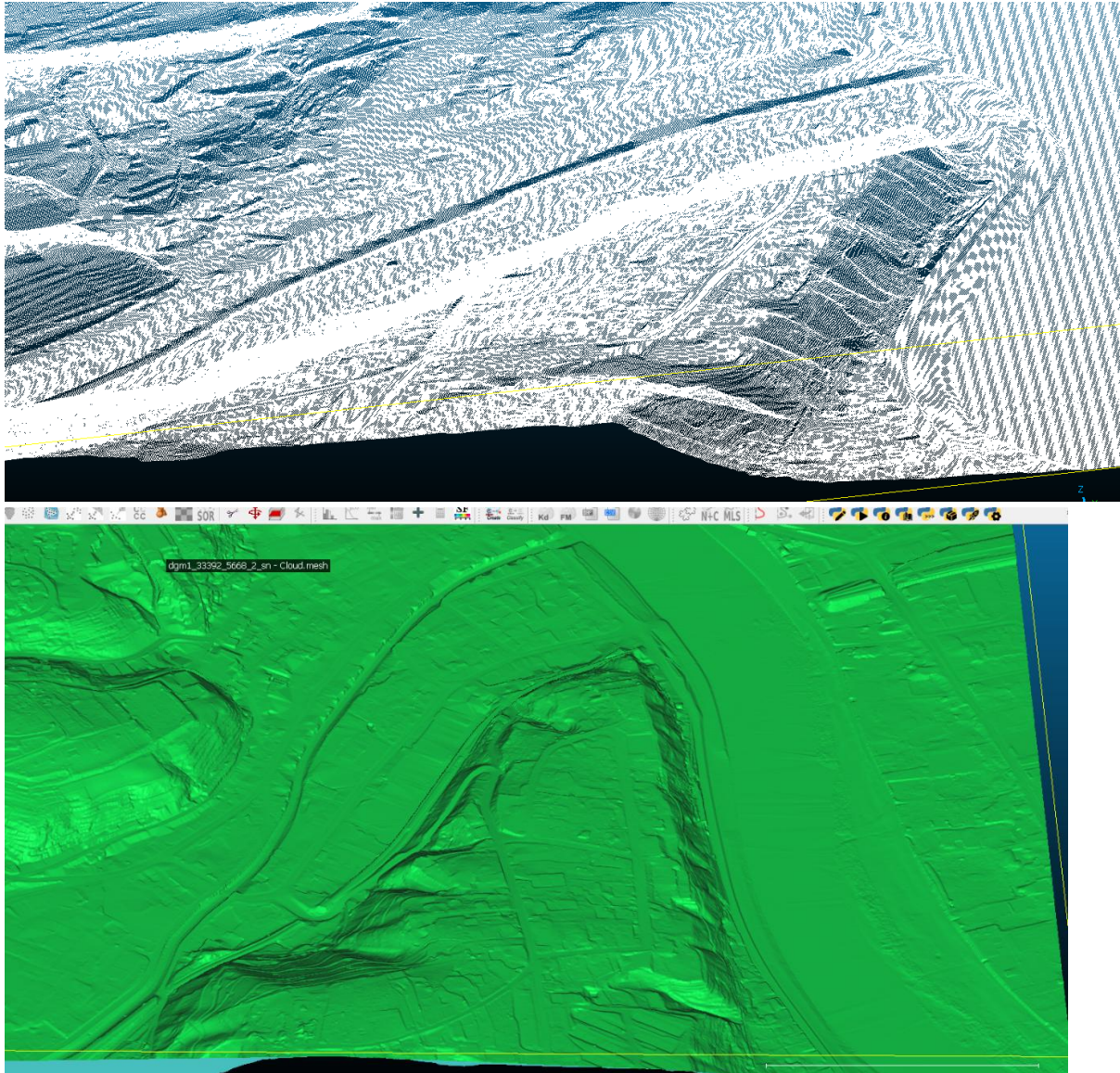


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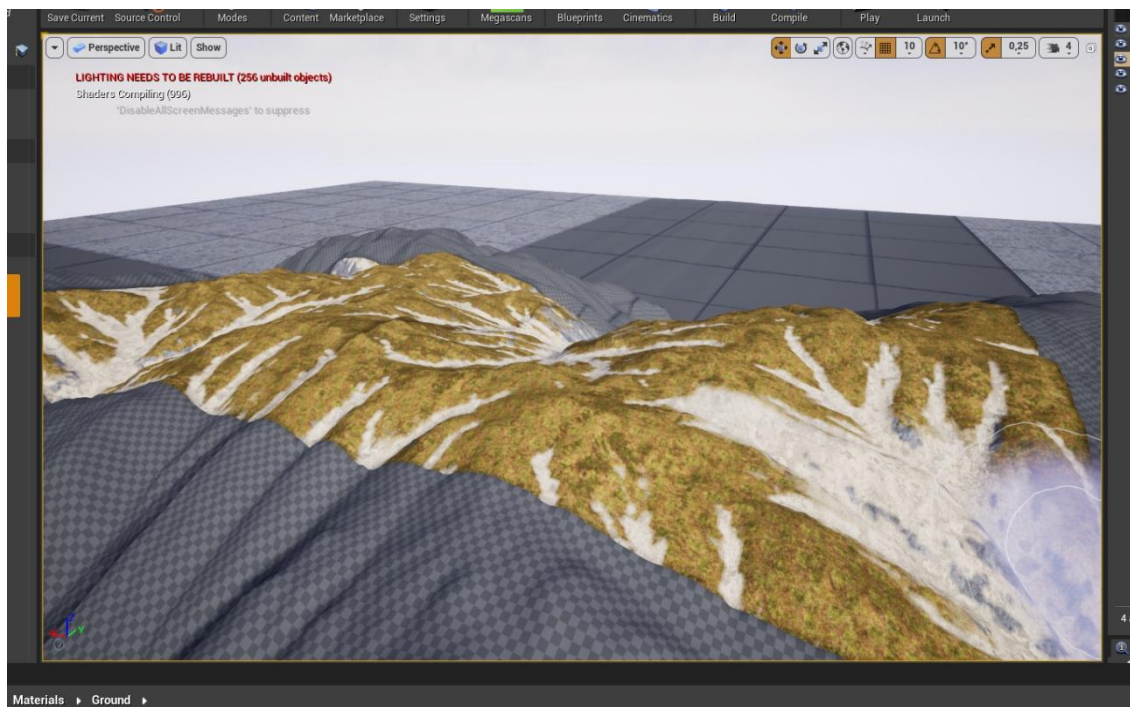
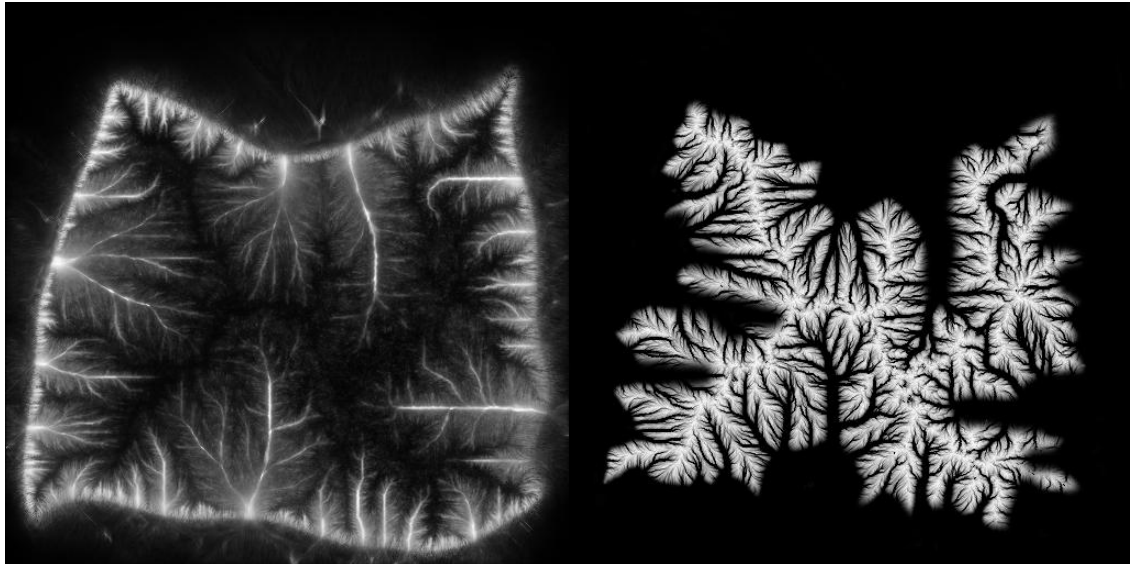
- CloudCompare
- <https://www.danielgm.net/cc/>
- <https://www.youtube.com/watch?v=jEQ93GIEJY8>



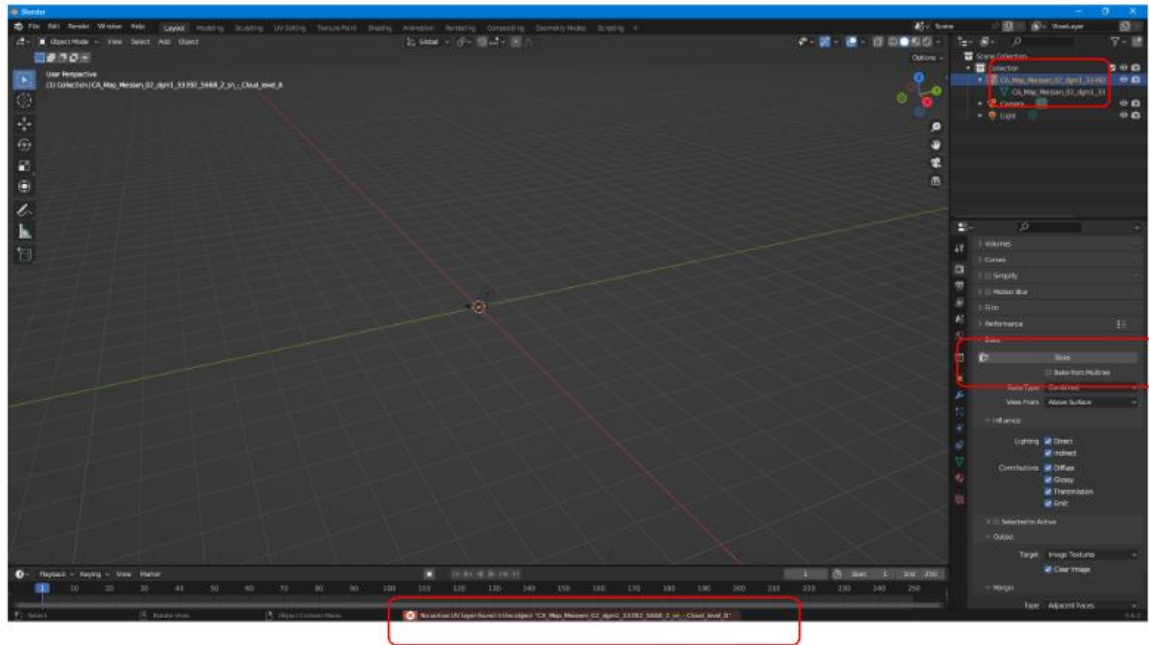


Setting up different layers for the same height map
(i didnt get this to work yet)

- Multilayer example from Zachary,
a picture per layer.
- I want only height info, layers are auto-created!
☐ one layer is sufficient!



BLENDER BAKE



- <https://devforum.roblox.com/t/using-blender-to-generate-properly-scaled-heightmaps-from-a-mesh/382209>
- Imported mesh into blender
- 1. and there you can click on "BAKE"
- 2. but i got the error of "no active UV layer found in the object"

using Geodata and displacement in Blender

- [HTTPS://Heightmap.Skydark.PI/](https://Heightmap.Skydark.PI/)
- <https://youtu.be/uHFSAnPgsEA?si=1MOq1Jw2wmqhWoMC>
- PNG download
- Blender:
 - renderer Cycles, Auflösung runter
 - Create plane
 - Subdivide
 - Shading/ plane /neues material (hat displacement) strg+T
 - Ins colour rein /material einstellungen /displacement and bump
 - Skalierung anpassen

TerrainMagic

- <https://www.youtube.com/watch?v=HD2OIIUOmLA>
-

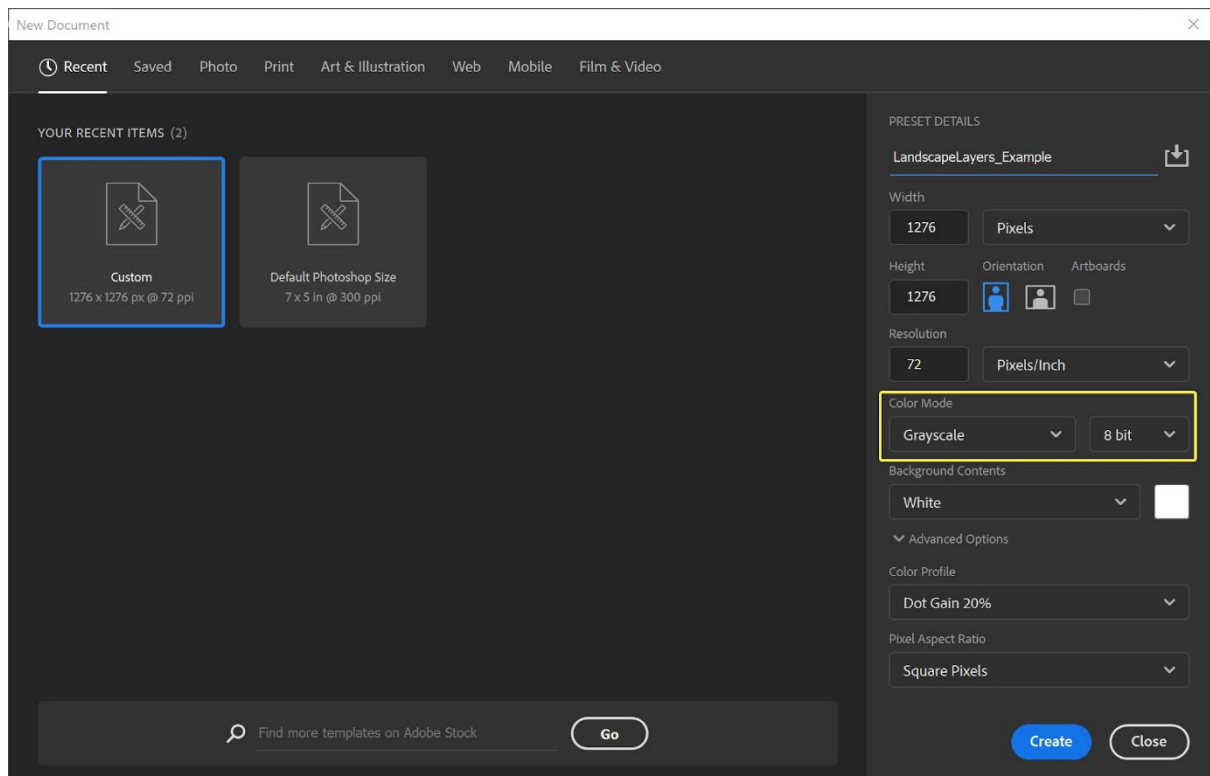
1.3 Import Heightmap Data into UNREAL

<https://docs.unrealengine.com/4.27/en-US/BuildingWorlds/Landscape/aq/>

- <https://docs.unrealengine.com/4.27/en-US/BuildingWorlds/Landscape/Custom/>
„Landscape layers can be imported from external programs via implementation of the ILandscapeHeightmapFileFormat and ILandscapeWeightmapFileFormat interfaces. The editor's existing support for image-based importing has been converted to use this interface and is fully supported. Images for the built-in formats are still required to be

grayscale, 8 bits per pixel, single channel files in either .PNG or .RAW format. If you are creating layers in Photoshop, use the following settings when creating a new document:“

- **Heightmap Formats:** When exporting heightmaps from external programs, only the following formats are usable inside UE4. 16-bit, grayscale PNG file 16-bit, grayscale RAW file in little-endian byte order



2. Polish Heightmap

height edits: create reasonable planes, valleys and mountains

auto material tweaks: make sure transition slopes between materials are good

create a visual barrier if possible

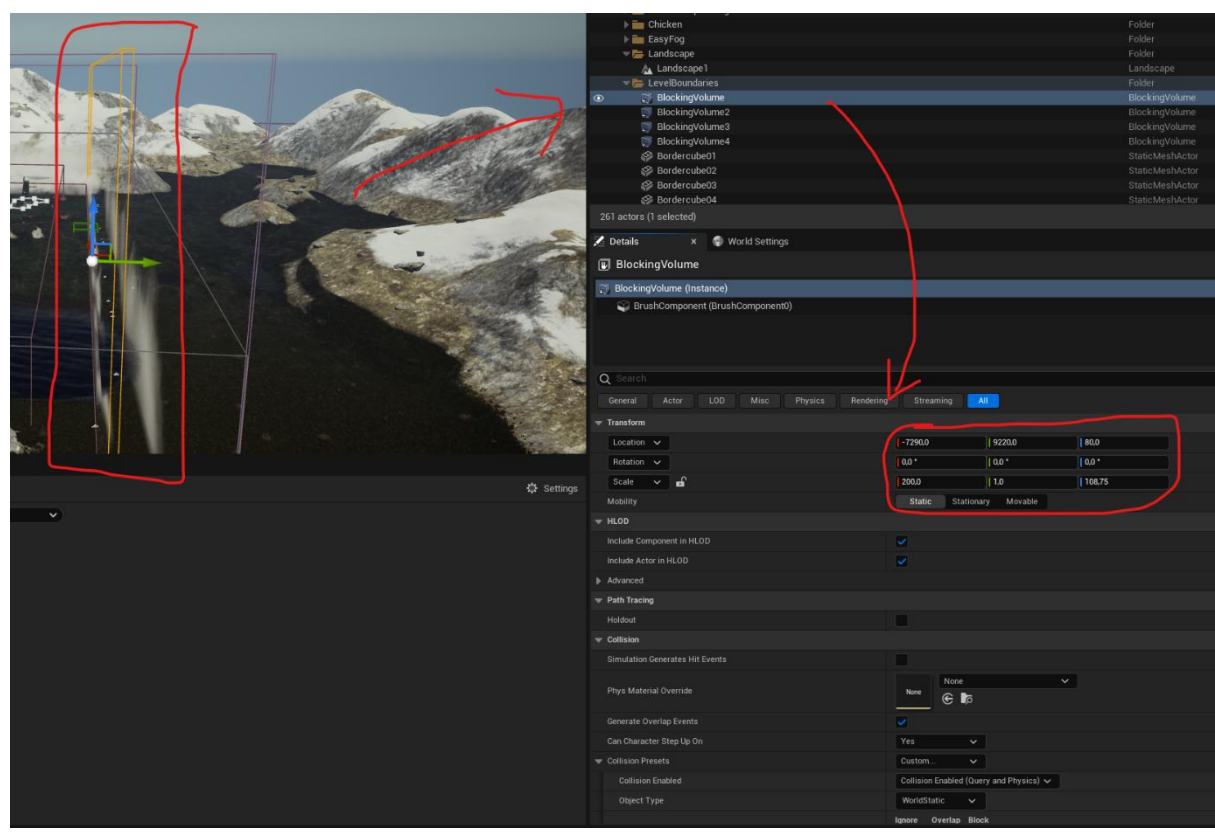
3. Level Boundaries

don't let the camera or units run out!

Use a different perspective in viewport also helps.



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Search

General

Actor

LOD

Misc

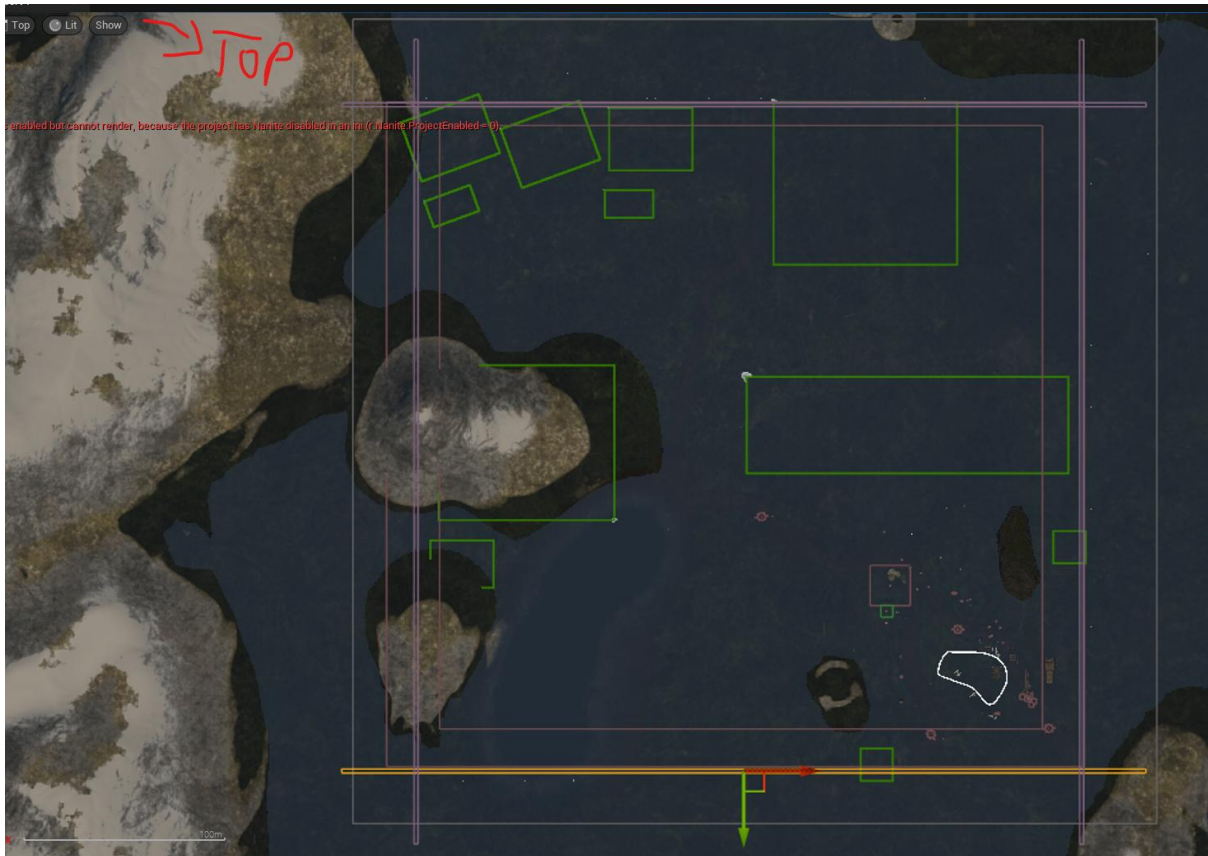
Physics

Rendering

Streaming

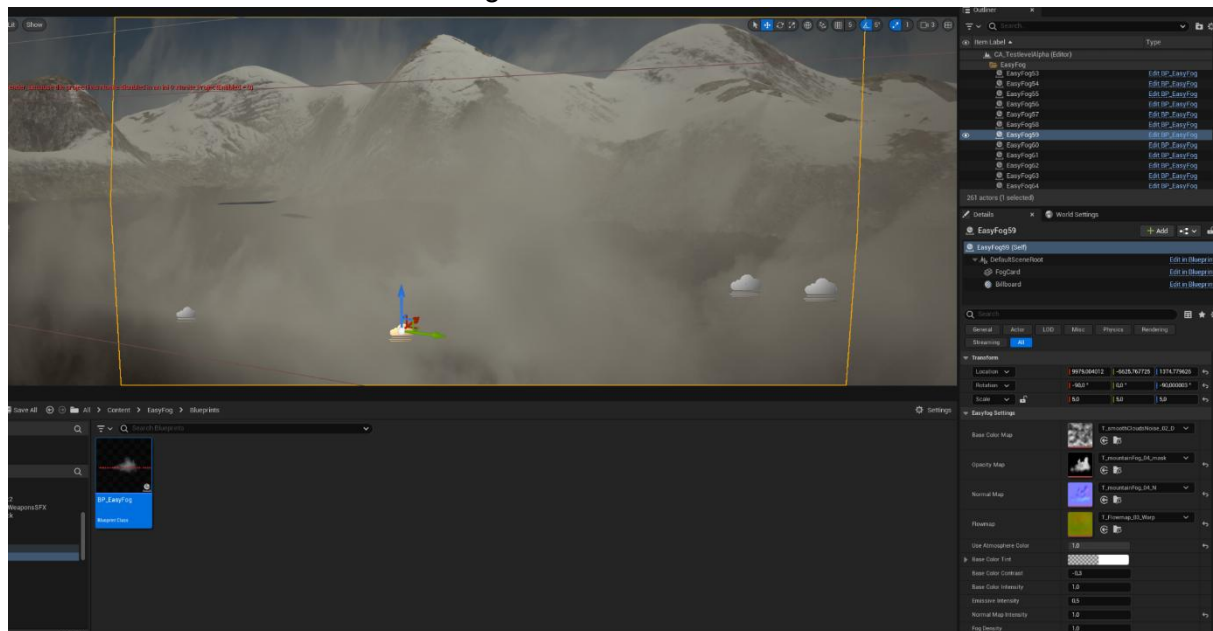
All

	Ignore	Overlap	Block
Collision Responses ?	☐	☐	☐
Trace Responses			
Visibility	☒	☐	☐
Camera	☐	☐	☒
Triggerbox	☒	☐	☐
Object Responses			
WorldStatic	☐	☐	☒
WorldDynamic	☐	☐	☒
Pawn	☐	☐	☒
PhysicsBody	☐	☐	☒
Vehicle	☐	☐	☒
Destructible	☐	☐	☒
CA_Landscape	☐	☐	☒
MapBoundaries	☐	☐	☒
SmallProjectiles	☐	☐	☒
LargeProjectiles	☐	☐	☒
Generate Overlap Events During Level Streaming	☐		
Update Overlaps Method During Level Streaming	Use Config Default ▾		
Default Update Overlaps Method During Level Streaming	Only Update Movable ▾		



4. World Fog

additional visual barrier to the sides. gentle but effectful.

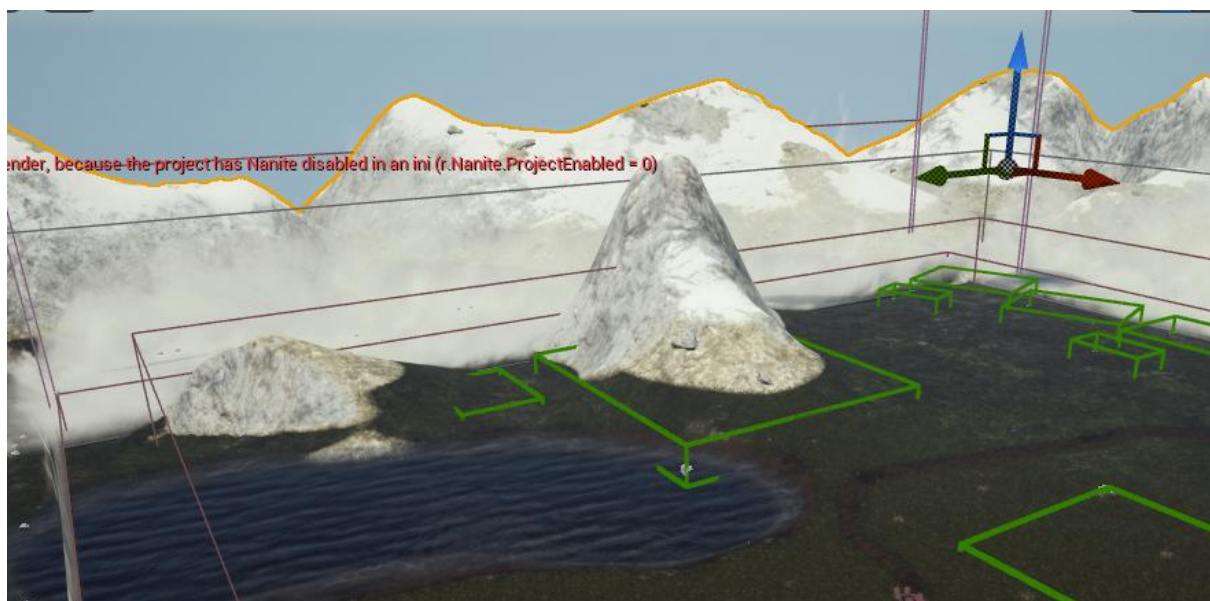




5. Apply Auto Material

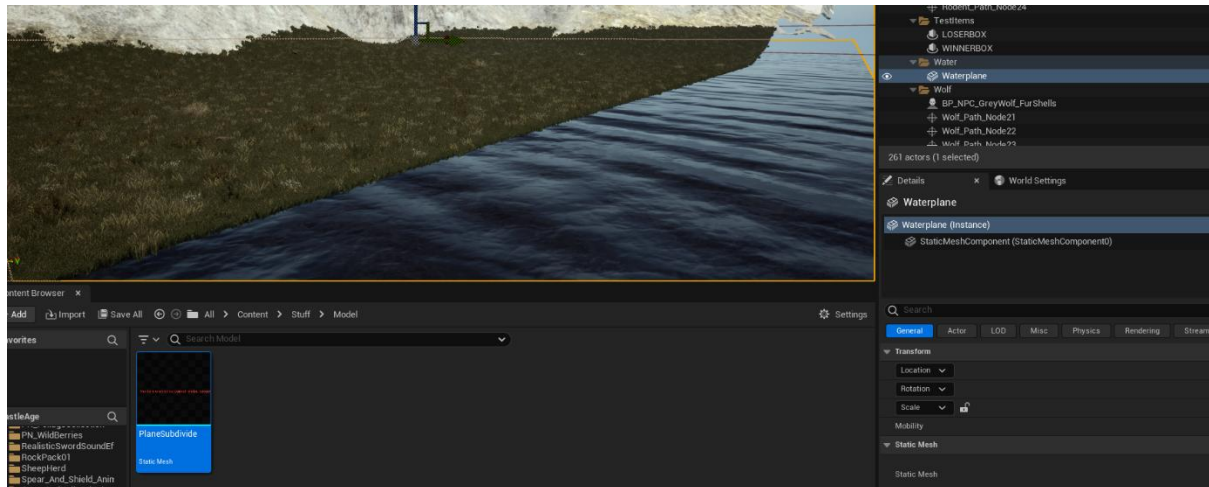
Now we can see how material transition slopes happen. Perhaps some manual painting for specific tactical placement or historical places is necessary.

6. Place Trees



7. Place Rock Boulders

8. Place Water Plane underneath Landscape



9. Create Navmesh

Steps to Generate NavMesh

1. Add Volume: Go to the Place Actors panel (Window > Place Actors), search for "Nav Mesh Bounds Volume", and drag it into the level.
2. Scale Volume: Scale the volume to encompass all areas where AI should walk.
3. Visualize: Press the P key on your keyboard to toggle the green visualization of the NavMesh.
4. Automatic Build: Unreal automatically generates (bakes) the mesh within this volume immediately upon adding it.

This video demonstrates how to add and scale a NavMesh Bounds Volume:

[ZenvaYouTube• 20.11.2022](#)

Key Configuration Settings (Project Settings > Navigation Mesh)

- Agent Radius/Height: Defines the size of the AI characters.
- Agent Max Slope: Sets the maximum steepness the AI can walk.
- Max Step Height: Dictates the height of obstacles the AI can step over.
- Runtime Generation: Set to "Dynamic" to update the navmesh during gameplay, or "Static" for bake-only.

You can watch this video to see how to configure the NavMesh settings:

[underscoreYouTube• 17.03.2022](#)

Pro Tips

- Dynamic Navigation: For procedural levels, set Runtime Generation to "Dynamic" and use Navigation Invokers to generate mesh around players only.
- Modifiers: Use Nav Modifier Volumes to mark areas as dangerous or blocked.
- Auto Link Generation: In UE 5.5+, you can enable "generate nav links" in the Recast NavMesh settings to automatically create jumping/falling links for AI.



•
This video explains how to create dynamic navmesh with navigation invokers:
[GamingDev2020YouTube• 04.09.2023](https://www.youtube.com/watch?v=ENMuqbNSLTs)

10. Tweak UltraDynamic Skies & Weather

<https://www.youtube.com/watch?v=ENMuqbNSLTs>

Ultra Dynamic Sky - Product Video + Quick Start (Newest UE5 Version)

11. Placing Encounter Animals

- place encounter units or spawners
- place walking nodes
- confirm that enemy tags and so on are correctly set up

12. Placing Player Start Camps

- place encounter units or spawners

13. Placing NPC camps

- place encounter units or spawners

14. Placing traveler spawners

- place encounter units or spawners

15. Setting up Birds

- place BP_CrowSpline in level
- place NS_SplineFlock in level
- move the Flock in object window under the spline
- adjust the settings for the Flock, adjust the spline points in the spline



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Go to : Content/FlockSystem/Blueprints/BP_CrowSpline

Drag To Level → Drag and drop the BP system on desired area. Set Spline → Drag and drop spline points.

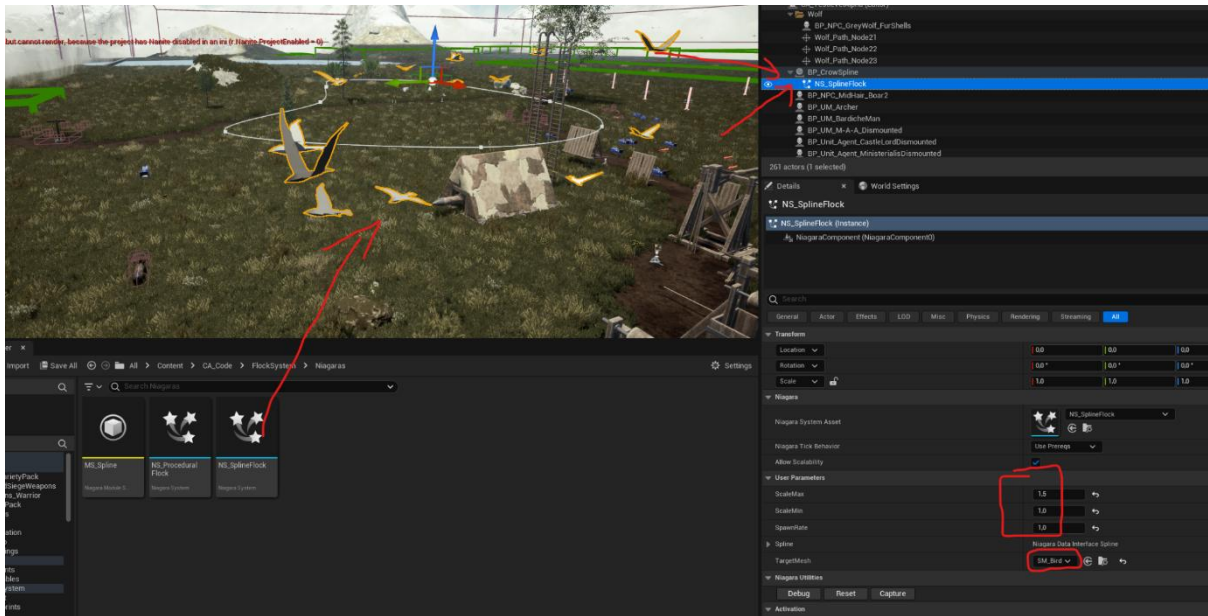
Go to : Content/FlockSystem/Niagaras/NS_SplineFlock

Add NiagaraFX → Add NS_SplineFlock on the current level

Drag For Spline Initialization → Drag NS_SplineFlock object under BP_CrowSpline

Reset Location → Set child NS_SplineFlock location to (0,0,0)

Quick Set NiagaraFX → Set Spawn Rate,Scale sizes, and select a static mesh



Procedural Parameters

Go to : Content/FlockSystem/Niagaras/NS_ProceduralFlock or NS_SplineFlock

COLOR Mixes the color with randomly selected percentage "max" and "min" values.

CURL NOISE FORCE Value of creating clutter along the flow axis

DRAG Drag value for per particle

DRIVE DIRECTION Particle movement direction

POINT FORCE Force for pursuit movement direction

SCALE Spawn the meshes with randomly selected percentage "max" and "min" values.

SPAWN RATE Augment or diminue particle count of simulation

TARGET MESH Niagara particle mesh

VORTEX FORCE Force for spiral effect



Go to : `Content/FlockSystem/Asset/Materials/M_UVMoverBaseBird`

Vertex Animation: Moving/Offset UV Coord World Position

`UV COORD` moving texture coordinates with random values on the target axis

Add

Go to : `Content/FlockSystem/Asset/Crow/Crow_VFX`

User can add custom mesh with the examine the sample Crow_VFX static mesh. The big step is separate and place only the moveable part in to UV0

Set up Level Blueprint

we will try to handle this somewhere else.